

NARSIMHA REDDY ENGINEERING COLLEGE

(UGC - AUTONOMOUS INSTITUTION • APPROVED BY AICTE, NEW DELHI • AFFILIATED TO JNTUH)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

MAJOR CAPSTONE PROJECT PRESENTATION • ACADEMIC YEAR 2026-2027

PROJECT CODE: BPY-CSE-2666 • CLINICAL DECISION SUPPORT SYSTEM (CDSS)

HealthNova AI: Patient Risk Level Prediction System

Explainable Clinical Acuity Stratification with Multi-Class ML & TreeSHAP Attribution

Continuous bedside telemetry monitoring combining Daphne ASGI real-time channels with calibrated Random Forest inference to prevent emergency department & ICU deterioration. Features sub-millisecond CPU inference (0.136ms), 0.985 ROC-AUC, 0.0027 calibrated Brier score, and sub-12ms game-theoretic TreeSHAP explainability.

Core Stack: Python 3.13 • Django 5 ASGI • Daphne • Redis 7 • Next.js 16.3 • React 19 • Neon PostgreSQL 16 • pgvector • Scikit-Learn • TreeSHAP

PROJECT PRESENTATION TEAM (DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING)

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1) Introduction — Problem & Solution

Why Traditional Scores Fail & How HealthNova AI Solves It

💡 **Key Takeaway:** Continuous, explainable AI to detect cardiac deterioration in real time.

POINT 01

Traditional Scores Fail

- ✓ **24-Hour Delay:** TIMI & APACHE II update only once daily
- ✓ **Misses Acute Crashes:** ICU patients deteriorate in minutes
- ✓ **Static Paper Reports:** Cannot process continuous live vitals

Problem: 24h Static Delay

POINT 02

Alarm Fatigue & Black Box

- ✓ **85%+ False Alarms:** Causes severe nurse burnout and fatigue
- ✓ **Black-Box Deep AI:** Neural networks hide the reason 'why'
- ✓ **Clinician Distrust:** Doctors reject unexplained predictions

Barrier: False Alarms & Black Box

POINT 03

The HealthNova Solution

- ✓ **Real-Time 4-Tier Risk:** Continuous Low, Mod, High, Critical
- ✓ **Sub-12ms XAI:** TreeSHAP explains every biomarker factor
- ✓ **100% Doctor in the Loop:** Hard medical fail-safes override AI

Solution: Real-Time Explainable CDSS

2) Architecture of Our Project

Decoupled Dual-Stack Foundation

💡 Key Takeaway: High-speed Python ASGI streaming paired with Next.js 16 and Neon PostgreSQL.

POINT 01

Python ASGI Backend

- ✓ **Python 3.13 & Django 5:** Core asynchronous clinical logic
- ✓ **Daphne & Redis 7:** Live WebSockets with < 20ms latency
- ✓ **Celery Task Queues:** Non-blocking background analytics

Backend: Django ASGI • Daphne • Redis 7

POINT 02

Neon PostgreSQL 16

- ✓ **Authoritative Database:** Single trusted source for all data
- ✓ **pgvector Extension:** Instant semantic protocol retrieval
- ✓ **Zero-Trust Security:** Complete zero-PHI privacy isolation

Database: Neon PostgreSQL 16 • pgvector

POINT 03

Next.js 16 Workspaces

- ✓ **Next.js 16 & React 19:** 173 fast compiled hospital routes
- ✓ **120 FPS SVG Canvas:** Fluid bedside ECG waveform display
- ✓ **Sub-250ms Load Times:** Zero external tracking for privacy

Frontend: Next.js 16.3 • React 19

3) Working Process — Real-Time Pipeline

From Raw Bedside Vitals to Calibrated Risk & TreeSHAP Attribution

💡 Key Takeaway: Live vitals stream into calibrated Random Forest (0.136ms CPU) and TreeSHAP explains drivers.

POINT 01

1. Live Ingestion

✓ **Live Vitals Stream:** Ingests HR, BP, SpO2, and GCS signals

✓ **Data Cleaning:** Outlier filtering & z-score normalization

✓ **Instant Vector:** 14 clinical features prepped in < 0.1ms

Step 1: Live Ingestion & Cleaning

POINT 02

2. Calibrated ML Inference

✓ **Random Forest (150 Trees):** Champion 0.985 ROC-AUC score

✓ **Platt Scaling:** True probabilities (0.0027 Brier error)

✓ **0.136ms CPU Speed:** Ultra-fast with zero GPU cost

Step 2: 0.136ms CPU ML Prediction

POINT 03

3. TreeSHAP & Safety

✓ **TreeSHAP in < 12ms:** Breaks down exact biomarker hazard drivers

✓ **Emergency Overrides:** qSOFA ≥ 2 & NEWS2 ≥ 5 trigger alerts

✓ **Uncertainty Guard:** Flags review if entropy > 0.65

Step 3: TreeSHAP XAI & Overrides

4) Technologies — Core Stack

Modern Full-Stack Ecosystem

💡 **Key Takeaway:** Production-grade stack built for sub-second performance, explainability, and security.

POINT 01

Backend & Streaming

- ✓ **Python 3.13:** Core high-performance programming runtime
- ✓ **Django 5 & Daphne:** ASGI real-time WebSocket server
- ✓ **Redis 7 & Celery:** In-memory pub/sub and task workers

Backend: Python 3.13 • Daphne • Redis 7

POINT 02

AI & Explainability

- ✓ **Scikit-Learn:** 150-tree Random Forest classifier engine
- ✓ **Platt Calibration:** Multi-class probability scaling module
- ✓ **TreeSHAP (shap):** Game-theoretic feature attribution

AI/ML: Scikit-Learn • TreeSHAP • Platt

POINT 03

Frontend, DB & DevOps

- ✓ **Next.js 16.3 & React 19:** Turbopack SSR/CSR workspaces
- ✓ **Neon PostgreSQL 16:** Serverless cloud DB with pgvector
- ✓ **Docker Microservices:** Multi-stage container builds (< 180MB)

Client/DB: Next.js 16 • Neon PG • Docker

5) Workflow — Care Pathway

From Emergency Intake to Doctor Review & MLOps Audit

💡 Key Takeaway: Closed-loop care pathway connecting Nurse Triage, AI Ingestion, Doctor Review, and MLOps.

POINT 01

Phase 1: Nurse Triage

- ✓ **< 45s Vital Intake:** Rapid bedside data entry (/nurse)
- ✓ **Automated ESI Triage:** Immediate 5-level acuity assignment
- ✓ **Direct Live Push:** Streams vitals to ICU monitor grid

Phase 1: Bedside Nurse Intake

POINT 02

Phase 2: AI Evaluation

- ✓ **Zero-PHI Minimizer:** Strips patient identity for privacy
- ✓ **Sub-Millisecond Risk:** 4-tier acuity calculated in 0.136ms
- ✓ **Safety Rule Engine:** Deterministic qSOFA & NEWS2 check

Phase 2: Swarm Ingestion & Evaluation

POINT 03

Phase 3: Doctor & MLOps

- ✓ **ICU Ward Grid (/doctor):** Attending doctor reviews live ECG
- ✓ **100% Human Sign-Off:** Mandatory clinician verification
- ✓ **MLOps Drift Audit:** Continuous PSI < 0.1 covariate monitoring

Phase 3: Human Sign-Off & Audit

6) Conclusion — Key Results & Scope

Validated Breakthroughs, Doctor Sovereignty & Next Steps

💡 Key Takeaway: HealthNova AI proves that fast, calibrated, explainable AI empowers doctors to save lives.

POINT 01

Key Achievements

- ✓ **Champion 0.985 ROC-AUC:** High precision on acute crises
- ✓ **0.136ms CPU Latency:** Instant risk recalculation bedside
- ✓ **Zero Alert Fatigue:** Platt calibration eliminates false alarms

Results: 0.985 ROC-AUC • 0.136ms Speed

POINT 02

Safety & Doctor Control

- ✓ **100% Doctor Authority:** Zero autonomous prescriptions
- ✓ **Deterministic Overrides:** Hard rules protect crashing patients
- ✓ **FDA 21 CFR Part 11:** Tamper-proof immutable audit ledger

Safety: 100% Clinician Sovereignty

POINT 03

Future Roadmap

- ✓ **Multimodal AI:** Ingest raw 12-lead ECG & DICOM image scans
- ✓ **Federated Learning:** Multi-hospital collaborative training
- ✓ **Edge Deployment:** Bedside micro-controllers for rural clinics

Future: Multimodal & Federated Edge